Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

My program is going to contain loops, lists, and dataclasses to help me map out my code.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

The easy parts would be creating a movie list and developing the hints to go along with it. Some of the hardest part would be checking the answer and creating the blurry image

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

All the movies I am going to have should be implemented first and the parts that will take the longest is checking the answers to see when it is right.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

There might be other work I have to get done and I will use some Ai to help develop my ideas about my website.

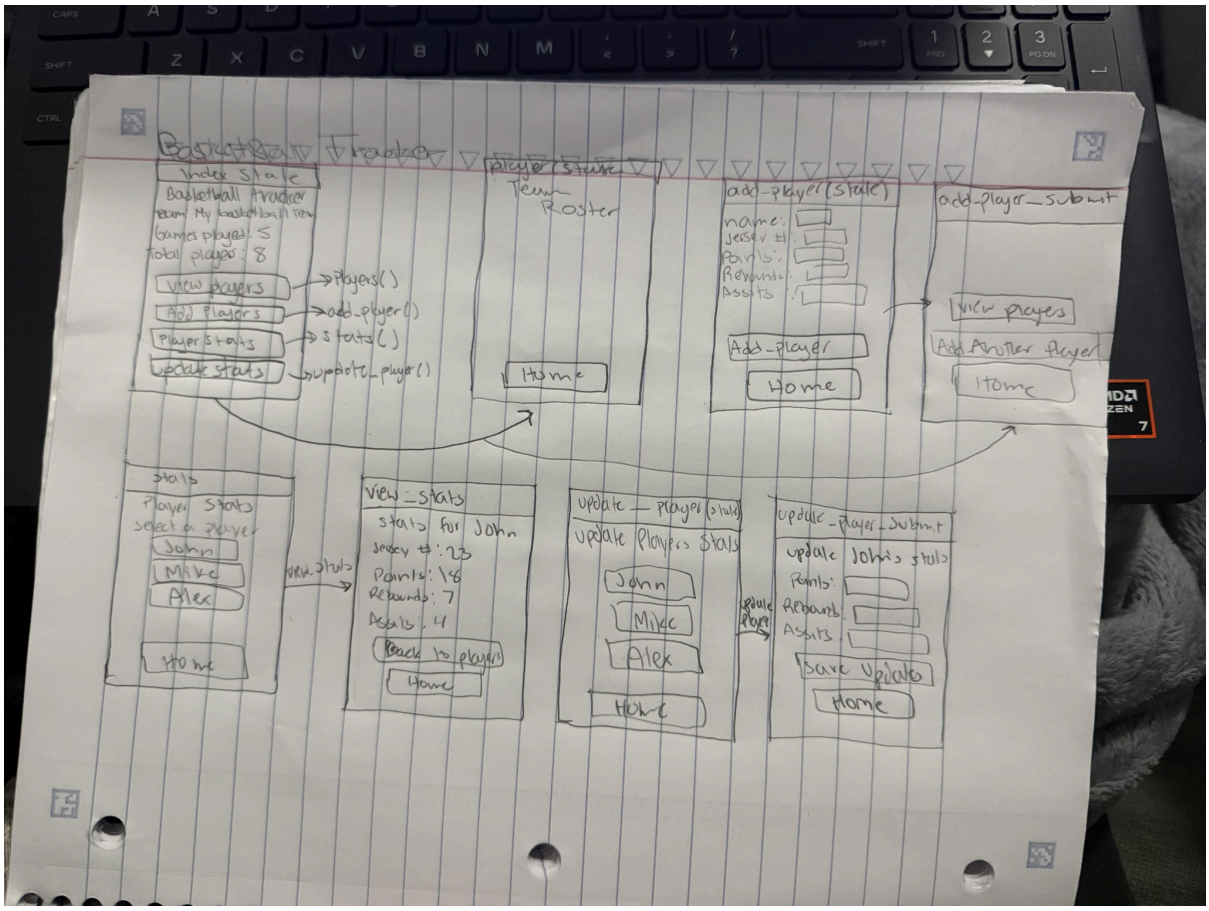
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.


Basketball Tracker
Index State
Basketball Tracker
Team: My basketball team
Games played: 5
Total players: 8
View players
Add Players
Player Stats
Update Stats
players()
add_player()
stats()
update_player()
player_state
Team Roster
Home
add_player(state)
name:
Jersey #:
Points:
Rebounds:
Assists:
Add_player
Home
add_player_submit
View players
Add Another player
Home
stats
Player Stats
Select a player
John
Mike
Alex
Home
view_stats
view_stats
Stats for John
Jersey #: 23
Points: 18
Rebounds: 7
Assists: 4
Back to players
Home
update_players(state)
update Players Stats
John
Mike
Alex
Home
update
update_player_submit
update John's stats
Points:
Rebounds:
Assists:
Save Updates
Home

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	**The point of the game is to show a blurry image and have the user guess the movie. If they do not get it on try one they can get a hint of the movie to help them.**
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	@dataclass Class Player : name: str jersey_number: int points: int rebounds: int assists: int Name would be the players name, points is how much they scored, rebounds is how many rebounds they got, assist is how many times they passed it to someone and they scored, Jersey_number is what number they have on their jersey.
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	Class State: 1. players: list[Player] - Stores a list of all basketball players. - Each player is an instance of the Player dataclass. - Starts as an empty list. - This allows us to add multiple players to the team. 2. team_name: str

	- Stores the name of the basketball team. - Starts as "My Basketball Team". - We can change it later. 3. games_played: int - Keeps track of how many games the team has played. - Starts at 0. - We can increase it after each game. 4. selected_player: str - Stores the name of the player currently selected. - Starts as an empty string. - Helps us know which player's statistics we want to update.
Routes (minimum 7) The names of your route functions	Route — Function / — Homepage and team overview /players — View all players /add-player — Add a new player /stats — View player statistics /update-player — Update player stats /leaderboard — Rank players by points /delete-player — Remove a player
Pages (minimum 5) A brief description of the pages in your site	Home Page – Displays the team name, games played, total players, and navigation buttons. Team Roster Page – Shows all players on the team, including their jersey numbers, points, rebounds, and assists.

	Add Player Page – Allows users to enter a player's name, jersey number, points, rebounds, and assists. Player Statistics Page – Lets users select a player and view their individual statistics. Update Player Stats Page – Lets users choose a player to update their statistics.
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	There will be text boxes that the user will type in for the code to obtain the information. And there will be boxes to click on to bring you from one page to another
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	Some of the statements will check if there are any players on the team, check if a players name is empty, check if jersey and points are in numbers, check if there are no players for statistics, and more
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it	For player in state.player: This loop will check through every player on the team.

is meaningful to the site	

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

Replace this text with your answer.

what are the 5 pages of the code

Your basketball tracker code is organized into **5 main sections (pages)**:

1. **Home Page** – Displays the team name, games played, total players, and navigation buttons.
2. **Team Roster Page** – Shows all players on the team, including their jersey numbers, points, rebounds, and assists.
3. **Add Player Page** – Allows users to enter a player's name, jersey number, points, rebounds, and assists.
4. **Player Statistics Page** – Lets users select a player and view their individual statistics.
5. **Update Player Stats Page** – Lets users choose a player to update their statistics.

Important: Your actual code has more than 5 pages. It also includes a leaderboard, delete player, and team settings pages.

what are my if statements

what are the if statements in the code i presented above

Your basketball tracker code has **10** `if` **statements:**

Line	If statement	What it does
47	`if len(state.players) == 0:`	Checks if there are no players on the team.
87	`if name.strip() == "":`	Checks if the player name is empty.
93	`if not jersey.isdigit() or not points.isdigit():`	Checks if the jersey number and points are numbers.
99	`if not rebounds.isdigit() or not assists.isdigit():`	Checks if rebounds and assists are numbers.
130	`if len(state.players) == 0:`	Checks if there are no players to view statistics for.
151	`if player.name == name:`	Checks if the selected player matches.
180	`if len(state.players) == 0:`	Checks if there are no players to update.
205	`if len(state.players) == 0:`	Checks if there are no players for statistics.
232	`if len(state.players) == 0:`	Checks if there are no players for the leaderboard.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.